

3	(a) (i)	resultant force on frame is zero <u>and</u> resultant torque is zero about any point	B1
	(ii)	Triangle forms a closed loop <u>and</u> direction of arrows are correct angles of vectors are correct	B1 B1
	(b)	$\sin 50^\circ = \frac{W/2}{T}$ or equivalent substitute $W = 1.0 \times 9.81$, $T = 6.4 \text{ N}$	M1 A1
	(c)	$T < 10$, so $\frac{W}{2 \sin q} < 10$ (allow equal sign)	M1
		$q > \sin^{-1} \left(\frac{1.0 \times 9.81}{2 \times 10} \right)$ $= 29.37^\circ$ $= 30^\circ$ accept 29°	A1

4	(a) (i)	1	B1
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